ABN SEAL COLLEGE
COOCHBEHAR
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Introduction to Ecology

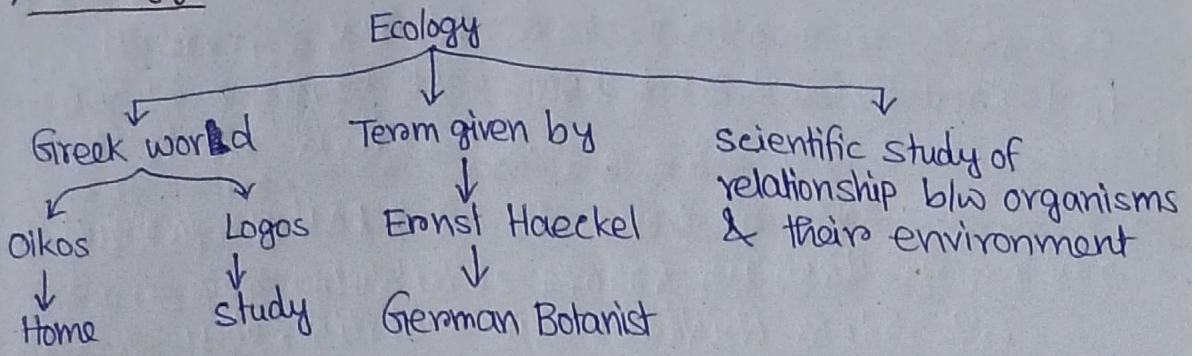


B.Sc. 2nd Semester

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1) Ecology



2) Level of organization

① Organism (Individual) ——> > A single living being is called organism.
> It is a basic unit of ecological study
> example - one tiger.

② Species ——> > A group of similar organism that can interbreed & produce fertile offspring is c/o species
> Example - All tiger belong to species Panthera tigris.

③ Population ——> > A group of organisms of same species ~~live~~ living in same area at the same time is c/o population.
> Example - population of tiger in a forest.

④ Community ——> > Different population living together & interacting ~~with~~ ~~each~~ in an area form a community.
> Example - Frog, bird, deer, tiger, insect living in a same forest.

⑤ Ecosystem ——> > A community together with its non-living environment (Biotic + Abiotic)
> Example - Forest ecosystem, pond ecosystem

⑥ Biome ——> > A large region with similar climate & vegetation.
> Example - Desert, grassland, Tundra biome

⑦ Biosphere ——> > The largest ecological level including all ecosystem of earth.
land, water & atm where life exists.

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3) Auto-ecology -

Autoecology is the branch of ecology that studies an individual organism or a single species in relation to its environment.

- It focuses on one species at a time
- It studies adaptation, behaviour, distribution, growth & life cycle of a species

Characteristics

1) studies individual species.
2) Simple & narrow approach.
3) Concerned ~~c~~ adaptation & environmental approach.
4) Experimental study is possible.

Example -

Study ~~on~~ of how cactus survive in desert conditions.

Synecology -

Synecology is the branch of ecology that studies group of organisms or communities & their interactions with the environment.

- It deals with many species together.
- It studies community structure, food chain, ecological succession & ~~adaptation~~ interactions.

Characteristics

1) Studies communities of organisms.
2) Broad & complex approach.
3) Concerned ~~c~~ interactions among species.
4) Studies ecosystem functioning.

Example -

Study of organisms in pond ecosystem & their relationships

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Autoecology	Synecology
1) Study of an individual organism or single species.	1) Study of communities of organisms.
2) Narrow & simple approach	2) Broad & Complex approach.
3) Focus on one species	3) Focus on many species together.
4) Studies adaptation & behaviour	4) Studies interaction among organism.
5) Experimental study is easier.	5) Experimental study is difficult.
6) Example- Study of cactus in desert.	6) Example- Study of pond ecosystem.

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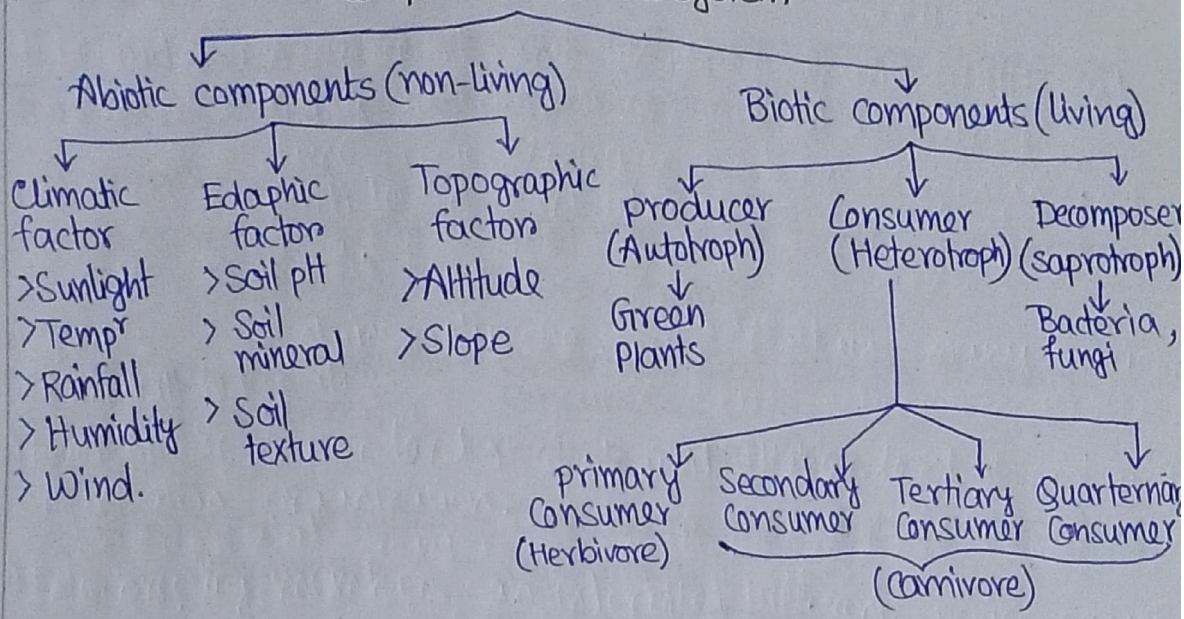
Biome	Biosphere
1) A large geographical area with ~~small~~ similar climate, vegetation & animals.	1) The global sum of all ecosystems where life exists.
2) Smaller unit of ecological organization.	2) Larger unit of ecological organization than biome.
3) Contains many ecosystem	3) Contain all biome & ecosystem.
4) Characterized mainly by climate & vegetation.	4) Includes all biotic factors & interactions ~~c~~ abiotic factor
5) Different types are ⊕	5) Only one biosphere esists on earth.
6) example- Desert biome, tundra biome	6) example- earth's biosphere
7) Biome is a part of Biosphere.	7) many ~~biosp~~ biomes togethe form biosphere.

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∅ Basis of Difference	Population	Community.
1) Defination	A group of ~~same species living~~ organisms of same species living in a ~~particle or~~ particular area.	A group of different population living & interacting in an area
2) Species composition	contains only one species	contains many species.
3) level of organization	lower level of ecological organization.	Higher level of ecological organization.
4) Interbreeding	members can interbreed	Different species cannot generally interbreed.
5) Nature of study	study characteristics of single species.	studies interactions among different species
6) Size	comparatively smaller unit	comparatively larger unit.
7) Complexity	Simpler	more complex
8) Interactions	mostly within same species	blw different species
9) Ecological focus	population density, Natality mortality, growth rate	Food chain, competition succession, symbiosis
10) Dependency	population can exist independently	Community depends on several populations
11) example	population of deer in a forest	Deer, tiger, grasses, bird & insects ect in a forest.

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ø Components of ecosystem



Producer (Autotroph)

producers are organisms that can prepare their own food from simple inorganic substances using sunlight or chemical energy.

ø Types -

① Photoautotrophs - use sunlight for food production
ex - green plants, algae

② Chemoautotrophs - use oxidation of chemicals/chemical energy for food production.
ex - Nitrifying bacteria.

ø Characteristics -

1) possess chlorophyll in most cases.
2) Form the first trophic level ~~in an~~ of a food chain.
3) Convert solar energy into chemical energy.
4) provide food & energy to consumers.
5) Essential for maintaining ecosystem balance.

ø Examples -

Grasses
Mango tree
Algae
Cyanobacteria.

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Consumer (Heterotroph)

Consumers are the organisms that cannot prepare their own food and depend on other organisms for nutrition.

ϕ Types-

① primary consumer- Organisms that feed on producers.
(Herbivores) ex- Deer, rabbit.

② Secondary consumer- Organisms that feed on primary consumers.
(carnivores) ex- Frog, small fish

③ Tertiary consumers- Organisms that feed on secondary consumers.
(carnivores) ex- Snake, Tiger.

④ Omnivores - Organisms that feed on both plants & animals.
ex - Human being, bear.

ϕ characteristics-

① cannot synthesize their own food.
② Depend directly/ indirectly on producer.
③ Occupy different trophic level.
④ Help in transfer of energy within the ecosystem.
⑤ may be herbivores, carnivores or omnivores,

ϕ Examples- Cow, Lion, human being, Eagle.

Decomposer (saprotroph)

Decomposer are the organisms that ~~el~~ breakdown dead & decaying organic matter into simpler substances.

ϕ Types-

① Microdecomposers - microscopic organisms
ex- bacteria.

② Macro decomposers - larger decomposing organisms.
ex- fungi, earthworm

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Characteristics-

① Feed on dead plants & animals.
② Convert complex organic matter into simple nutrients
③ Recycle nutrients back to the soil.
④ Maintain ecological balance.
⑤ Important in nutrient cycling.

Examples-

Bacteria, mushroom, fungi, earthworm

Homeostasis-

Homeostasis is the ability of an organism or ecosystem to maintain a stable internal environment despite changes in external environment despite changes in external environmental conditions.

Ø Types of feedback mechanism in Homeostasis

① +ve feedback-

+ve feedback is a mechanism in which the response increases or amplifies the original stimulus, causing the process to continue further.

ex- During childbirth, the release of oxytocin increases uterine contraction, which further stimulates more oxytocin release.

② -ve feedback-

-ve feedback is a mechanism in which the response opposes or reduces the original ~~stimuto~~ stimulus, helping to maintain stability & balance.

ex- When the body temperature increases, sweating occur to cool the body & restore normal temperature.

Ø Regulation in plants by Homeostasis-

Plants maintain internal balance through homeostasis by regulating different physiological processes. The main regulations are-

① Water balance regulation- Stomata open & close to control water loss ~~during~~ through transpiration.

② Temperature regulation- Transpiration cools the plant ~~body~~ body during high temperature.

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③ Gas exchange regulation- Stomata regulate the ~~exchang~~ exchange of oxygen & carbon dioxide.

④ Mineral balance regulation- Root absorb required minerals and maintain ionic balance.

⑤ Osmotic regulation- Vacuoles help maintain cell turgidity and water balance.

⑥ Hormonal regulation- plant hormone coordinate growth and response to environment.

⑦ Food storage & usage- Plant regulate synthesis & utilization of ~~resou~~ food according to need.

ØEcological importance of Homeostasis-

① Maintain balance in the ecosystem.
② Helps organisms survive under changing environmental conditions.
③ Regulates factors like temp^{n}, water balance & pH
④ Ensure proper functioning of metabolic activities.
⑤ Maintain stability of population & community structure.
⑥ prevent sudden ecological balance.
⑦ Support continuity & sustainability of life in the biosphere.

Ø Resistance

Resistance is the ability of an organism or ecosystem to withstand environmental disturbances without significant change.

Characteristics -

① prevent major changes in ecosystem structure.
② Helps maintain ecological stability.
③ Reduce the effect of disturbances.

example -

A forest remaining stable during a short draught.

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Resilience-

Resilience is the ability of an organism or ecosystem to recover quickly after a disturbance or damage.

Characteristics

① Helps ecosystem return to its normal condition.
② Indicates recovery capacity ~~of~~ of an organism.
③ Important for long term ecological balance.

example-

Grassland regrows after a forest fire or flood.

BOD-

Full Form- Biological Oxygen Demand/
Biochemical Oxygen Demand

BOD is the amount of dissolved O_2 required by aerobic microorganisms to decompose biodegradable organic matter present in water under suitable conditions of temperature & time.

> It is an important indicator of organic pollution in water.
> BOD is usually measured for 5 days at 20°C, called BOD_5
> When ~~Sw~~ sewage or organic waste enters water bodies ~~bec~~ bacteria decompose it & consume oxygen
> High BOD means a large amount of organic waste is present ~~in~~ & the water is highly polluted.
> Low BOD indicates cleaner water with less organic pollution.

Importance

1) Help determine water pollution level.
2) Used in wastewater treatment plants.
3) Indicates amount of biodegradable matter in water.
4) Helps assess suitability of water for aquatic life.

~~sewage~~ generally show high BOD.

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COD-

Full Form- Chemical Oxygen Demand.

COD is the ~~amount~~ amount of oxygen required to chemically oxydize both biodegradable & non-biodegradable organic substances present in water.

→ It measure the total quantity of pollutants in water.

→ COD test uses strong ~~oxy~~ oxidizing agents such as potassium dichromate in acidic medium.

→ High COD value indicates heavy pollution due to organic & chemical wastes.

→ It is faster than BOD & usually completed within a few hours.

Importance

① Measure total ~~BOD~~ pollution load in water.
② Useful for testing industrial wastewater.
③ Faster method for water quality analysis.
④ Helps monitor efficiency of treatment plants

ex- Industrial waste from textile, chemical & dye industries often have high COD.

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Basis of Difference	BOD	COD
1) Full Form	Biochemical Oxygen Demand or Biological Oxygen Demand	Chemical Oxygen Demand
2) Definition	Amount of oxygen required by aerobic microorganisms to decompose biodegradable organic matter present in water under suitable condition of temperature and light.	Amount of oxygen required to chemically decompose both biodegradable and nonbiodegradable organic matter present in water.
3) Type of process	Biological process.	chemical process.
4) Measures	Only biodegradable organic substances	Both biodegradable & non biodegradable organic substances & all oxidizable inorganic substances
5) Method used	use microorganisms	use chemical oxidizing agents.
6) Time required	About 5 days	Few hours
7) Accuracy	less comprehensive	more comprehensive
8) pollution indicated	Biological pollution	Total pollution load.
9) value relation	usually lower than COD	usually higher than BOD
10) Common use	Domestic sewage analysis	Industrial wastewater analysis.

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